

General Relativity

Metrics and Line Elements, Math.

$ds^2 = g_{ab} dx^a dx^b$

$g_{ab} = g_{ba} \rightarrow$ total components

$A^a = g^{ab} B_b, \langle \beta, A \rangle = \beta_a A^a$

$J^a_b = \frac{\partial x^a}{\partial x'^b}, (J^{-1})^b_a = \frac{\partial x'^b}{\partial x^a}$

$g'_{ab} = J^c_a J^d_b g_{cd} \Rightarrow g' = J^T g J$

$\det g' = (\det J)^2 \det g$

$A'^a = J^a_c A^c, \beta'_a = (J^{-1})^b_a \beta_b$

$T'^a_{\dots b} = J^a_c \dots J^d_{\dots b} (J^{-1})^c_d T^c_{\dots d}$

$A_a = g_{ab} A^b, A^a = g^{ab} A_b$

$\nabla_a g_{bc} = 0$

$\nabla_a V^b = \partial_a V^b + \Gamma^b_{ac} V^c$

$\nabla_a \beta_b = \partial_a \beta_b - \Gamma^c_{ab} \beta_c$

$\nabla_a u^b = \partial_a u^b + \Gamma^b_{ac} u^c$

$\Gamma^a_{bc} = \Gamma^a_{cb}$

$\Gamma^a_{bc} = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc})$

$T^a_{bc} = \frac{\partial^2 x^a}{\partial x^b \partial x^c} + \frac{\partial x^a}{\partial x^d} \frac{\partial^2 x^d}{\partial x^b \partial x^c}$

$T^a_{bc} = \partial_b \ln \sqrt{|\det g|}, \nabla_a V^a = \frac{1}{\sqrt{|\det g|}} \partial_a (\sqrt{|\det g|} V^a)$

$T^a_{bc} = g_{ad} \Gamma^d_{bc} = \frac{1}{2} (\partial_b g_{ac} + \partial_c g_{ab} - \partial_a g_{bc})$

$\{T\} = \frac{1}{\text{length}}, \{R_{ab}\} = \frac{1}{\text{length}^2}, \{R^a_{bcd}\} = \frac{1}{\text{length}^2}$

$R^d_{abc} V^c = -[\nabla_a, \nabla_b] V^d, R^d_{abc} = g^d_e R^e_{abc}$

$R^d_{abc} + R^d_{bac} + R^d_{cab} = 0$

$\nabla_e R^d_{abc} + \nabla_a R^d_{bec} + \nabla_b R^d_{cae} = 0$

$R^d_{abc} = \partial_b T^d_{ac} - \partial_a T^d_{bc} + T^e_{bc} T^d_{ae} - T^e_{ac} T^d_{be}$

$\text{space locally flat iff } R^d_{abc} = 0$

$g_{ab} \text{ some } n^a = \frac{\partial x^a}{\partial s} \text{ describing sep. ltr 2 paths}$

$\frac{d^2 n^a}{ds^2} + R^a_{bcd} n^b n^c n^d = 0$

$R_{ac} = R^b_{abc}$

Cons and Facts

$g_{ab} dx^a dx^b < 0$ timelike

$g_{ab} dx^a dx^b > 0$ spacelike

$A^a = g_{ab} A^b = 0$ null

for diag. metric $ds^2 = g_{00}(dx^0)^2 + g_{11}(dx^1)^2 + g_{22}(dx^2)^2 + g_{33}(dx^3)^2$

6 Area elems, 4 Vol elems. in M, only are:

$dV^{(4)} = \sqrt{(-g_{00}g_{11}g_{22}g_{33})} dx^0 dx^1 dx^2 dx^3$

$= \sqrt{-\det g} d^4 x$

Hypersurfaces: defined by $f(x^a) = 0$

$\partial_a f$ normal to all tangent vels on hypersurf.

timelike hypersurf $\hat{n} = \frac{\partial f}{\sqrt{g_{ab}(\partial_a f)(\partial_b f)}}$

$(J^{-1})^a_b T^b_{\dots c} = \dots$

null hypersurf $g^{ab}(\partial_a f)(\partial_b f) = 0$

Geodesics: $S = -m \int \sqrt{-g_{ab} \frac{dx^a}{d\lambda} \frac{dx^b}{d\lambda}} d\lambda$

$\frac{d^2 x^a}{d\tau^2} + \Gamma^a_{bc} \frac{dx^b}{d\tau} \frac{dx^c}{d\tau} = 0$

$\frac{du^a}{d\tau} + \Gamma^a_{bc} u^b u^c = 0$

$\frac{du^a}{d\lambda} = 0, u^a \neq 0, \frac{du^a}{d\lambda} = \frac{dx^a}{d\lambda} \frac{du^a}{dx^a} = u^b \nabla_b u^a = 0$

$\frac{du^a}{d\tau} = \nabla_u u^a, \frac{du^a}{d\lambda} = 0, \frac{du^a}{d\lambda} = \frac{dx^a}{d\lambda} \frac{du^a}{dx^a} = u^b \nabla_b u^a = 0$

Parallel Transport: moving vector in M.

$\nabla_u V^a = 0, \nabla_V V^a = 0, \frac{dV^a}{d\lambda} = 0$

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Schwarzschild Soln.

$ds^2 = -\alpha(r)(dx^0)^2 + \beta(r)(dx^1)^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$

$T^0_0 = T^1_1 = \frac{1}{2\alpha} \frac{d\alpha}{dr}, T^2_2 = T^3_3 = \frac{1}{2\beta} \frac{d\beta}{dr}$

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Killing Vectors

$(1, 0, 0, 0) = \xi^0$ time trans. inv.

$(0, 0, 0, 1) = \xi^3$ rot about z-axis inv.

$\xi = -\xi^0 u = (1 - \frac{2M}{r}) u^0, \xi = \xi^3 u = r^2 \sin^2\theta u^3$

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spacetime line element

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$$

trajectories arbitrarily parameterized

$$x^\mu = (x^0, x^1, x^2, x^3) \Rightarrow x^\mu(\lambda) = (x^0(\lambda), \dots)$$

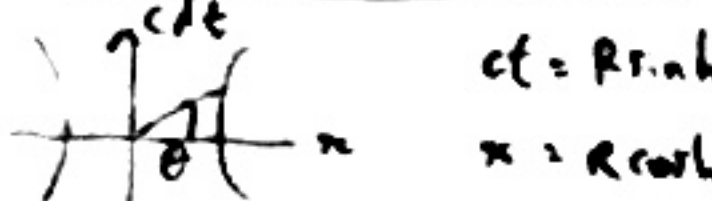
proper time interval

$$\tau = \int_A^B \sqrt{-ds^2} = \int_A^B \sqrt{-\left(\frac{dx^0}{d\lambda}\right)^2 + \left(\frac{dx^1}{d\lambda}\right)^2 + \dots}$$

defining proper time $d\tau^2 = -\frac{ds^2}{c^2}$

$$\tau_{AB} = \int_A^B \sqrt{dt^2 - \frac{1}{c^2}(dx^2 + dy^2 + dz^2)} = \int_A^B \sqrt{1 - \frac{v^2}{c^2}} dt$$

Angles/circles in spacetime



vector and coordinate laws

if general coords x^μ then

$$e_\mu = \frac{\partial}{\partial x^\mu} = \frac{\partial}{\partial x^\alpha} \frac{\partial x^\alpha}{\partial x^\mu}, g_{\mu\nu} = e_\mu \cdot e_\nu$$

curd basis e^μ, e^ν

$$e^\mu = \frac{\partial x^\mu}{\partial x^\alpha} e^\alpha \text{ so } v^\mu = \frac{\partial x^\mu}{\partial x^\alpha} v^\alpha$$

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$$g_{\mu\nu} = \frac{\partial x^\alpha}{\partial x^\mu} \frac{\partial x^\beta}{\partial x^\nu} g_{\alpha\beta}$$

rapidities

define $\eta = \tanh \phi$

$$\delta = \cosh \phi, \gamma v = \sinh \phi$$

$$\sqrt{1-v^2} = \frac{1}{\cosh \phi}$$

rapidity add linearly

$$\phi_{12} = \phi_1 + \phi_2$$

More vector laws

$$F^{\lambda}_{\mu} = F_{\mu}^{\lambda} \eta^{\lambda\alpha} \eta_{\alpha\mu} = \eta^{\alpha\lambda} F_{\alpha\mu} \eta_{\lambda\mu}$$

Kronecker Delta, Levi Civita

$$\delta^\mu_\nu = \eta^{\mu\alpha} \eta_{\alpha\nu}$$

$$\epsilon^{\mu\nu\rho\sigma} = \epsilon_{\mu\nu\rho\sigma}$$

$$F^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$$

$$(A \cdot B) = \epsilon_{\mu\nu\rho\sigma} A^\mu B^\nu$$

Black Hole Geometry

Edington-Finkelstein

$$d = x^0 + r = 2M \log \left(\frac{r}{2M} - 1 \right)$$

$$dN = \frac{dr}{r^2} = \frac{dx^0}{r^2}$$

$$ds^2 = -(1 - \frac{2M}{r}) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega^2$$

$$g_{\mu\nu} = \begin{pmatrix} -(1 - \frac{2M}{r}) & 0 & 0 & 0 \\ 0 & \frac{1}{1 - \frac{2M}{r}} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}$$

ingoing light: $v = \text{const}$

$$g^{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 - \frac{2M}{r} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2 \theta} \end{pmatrix}$$

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Time Dilation

$$\delta = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow \text{red. to } \frac{1}{\gamma}$$

Particle moves at v rel to frame S . If T is proper time, $t = \gamma T$ in frame S .

Length Contraction

object L_0 in rest frame. In frame moving at v , $L = L_0 / \gamma$

$$L = \frac{L_0}{\gamma}$$

Velocity Addition

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}}$$

For 2 Lorentz Transformations

$$\delta_{CA} = \delta_{CB} \delta_{BA} (1 + v_{BA} v_{CB})$$

Velocity Addition

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Four-vectors

$$a = (a^0, \vec{a}) = a^\mu e_\mu$$

$$a \cdot b = \eta_{\mu\nu} a^\mu b^\nu$$

$$\eta_{\mu\nu} = e_\mu \cdot e_\nu$$

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$$

Four-velocity

$$u^\mu = \frac{dx^\mu}{d\tau} = (\gamma, \gamma \vec{v})$$

$$u \cdot u = \eta_{\mu\nu} u^\mu u^\nu = -1$$

unit timelike, parallel to world line

Velocity Addition

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$$v' = \frac{v - u}{1 - \frac{uv}{c^2}}$$

For 2 Lorentz Transformations

$$\delta_{CA} = \delta_{CB} \delta_{BA} (1 + v_{BA} v_{CB})$$

Kinematic Laws

$$p^\mu = m u^\mu = (\gamma m, \gamma m \vec{v})$$

$$p = (E, \vec{p}) \text{ for free particles}$$

$$a^\mu = \frac{dp^\mu}{d\tau} = \frac{1}{m} \frac{dp^\mu}{d\tau}$$

$$\vec{a} \cdot \vec{v} = \frac{d\vec{p}}{dt} \cdot \vec{v} = \frac{dE}{dt}$$

$$f^\mu = (\delta F, \vec{v})$$

$$f^\mu f_\mu = m^2 a^\mu a_\mu = \text{frame-invariant}$$

conservation of four-momentum

$$p_{\text{kin}} = \text{constant}$$

$$p^\mu p_\mu = -m^2 \text{ Always}$$

Velocity Addition

$$v' = \frac{v - u}{1 - \frac{uv}{c^2}}$$

For 2 Lorentz Transformations

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Hydrodynamics

$$u^\mu = \frac{dx^\mu}{d\tau} = (\gamma, \gamma \vec{v})$$

$$p^\mu = m u^\mu = (\gamma m, \gamma m \vec{v})$$

$$p = (E, \vec{p}) \text{ for free particles}$$

$$a^\mu = \frac{dp^\mu}{d\tau} = \frac{1}{m} \frac{dp^\mu}{d\tau}$$

$$\vec{a} \cdot \vec{v} = \frac{d\vec{p}}{dt} \cdot \vec{v} = \frac{dE}{dt}$$

$$f^\mu = (\delta F, \vec{v})$$

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